

Divide Multi-Digit Numbers

Lesson 1-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In $156 \div 12 = 13$, the dividend is 156 — it is the total being split up

Dividend

The number you are splitting up in a division problem.

In $156 \div 12 = 13$, the divisor is 12 — it is the size of each group

Divisor

The number you split by in a division problem.

In $156 \div 12 = 13$, the quotient is 13 — there are 13 groups

Quotient

The answer when you divide.

$17 \div 5 = 3 R 2$ means 3 groups of 5 with 2 left over

Remainder

What is left over when a number does not divide evenly.

$1,344 \div 12$: first $12 \times 100 = 1,200$, then $12 \times 12 = 144$. Quotient = $100 + 12 = 112$

Partial quotients

A way to divide by breaking the problem into smaller, easier steps.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

1 Notice

The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation.

2 Key idea

I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

3 Words I'll use

Dividend

Divisor

Quotient

Remainder

Partial quotients



Think About It

- What total amount needs to be divided?
- How many groups are the supplies being split into?
- Will each section get the same amount with none left over?



Watch

Watch your teacher model one example. Jot what you see.



We try

Solve the next one together as a class.



You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The station must divide 1,344 nutrition bars equally among 12 crew sections. Before dividing, about how many bars should each section get, and how do you know?

Level 1 support

Start here: Estimating with friendly numbers checks whether the quotient is reasonable.

 **Need a hint?**

Sentence starters (optional)

Each section gets about ____ bars because 1,344 is close to ____.

Cada sección recibe unas ____ barras porque 1,344 está cerca de ____.

I estimated by ____.

Estimé al ____.

Word bank: dividend divisor quotient estimate partial quotients

Listen for: Students estimate using $1,200 \div 12 = 100$ (or similar) to predict roughly 100+ bars per section, identifying 1,344 as the dividend and 12 as the divisor.

Level 2

Will 1,344 divide evenly by 12, or will there be a remainder? Predict before computing and justify your reasoning.

- I predict the remainder is ____ because ____.
- 1,344 is divisible by 12 since ____.

EXPLORE

You used partial quotients for $1,344 \div 12$. What steps did you use, and how do they build to the final quotient?

Level 1 support

Start here: Partial quotients break a hard division into easy chunks you add together.

 **Need a hint?**

Sentence starters (optional)

First I took $12 \times \underline{\hspace{1cm}} = 1,200$, then $12 \times \underline{\hspace{1cm}} = 144$.

Primero usé $12 \times \underline{\hspace{1cm}} = 1,200$, luego $12 \times \underline{\hspace{1cm}} = 144$.

The quotient is $\underline{\hspace{1cm}}$ because I added the partial quotients $\underline{\hspace{1cm}}$.

El cociente es $\underline{\hspace{1cm}}$ porque sumé los cocientes parciales $\underline{\hspace{1cm}}$.

Word bank: **partial quotients** **dividend** **divisor** **quotient** **remainder**

Listen for: Students explain $12 \times 100 = 1,200$, then 144 left, $12 \times 12 = 144$, so the quotient is $100 + 12 = 112$ with no remainder.

Level 2

Why does it not matter whether you start with 12×100 or 12×50 in partial quotients?

Will you reach the same quotient?

- It does not matter where I start because $\underline{\hspace{1cm}}$.
- Different first chunks still give 112 since $\underline{\hspace{1cm}}$.

CONNECT

A school divides 1,680 pencils equally among 24 classrooms. How does multi-digit division tell each classroom's share, and how can you check it?

Level 1 support

Start here: Dividing 1,680 by 24 gives each classroom's count; multiply back to check.

 **Need a hint?**

Sentence starters (optional)

Each classroom gets ____ pencils because $1,680 \div 24 =$ ____.

Cada salón recibe ____ lápices porque $1,680 \div 24 =$ ____.

I can check by multiplying ____ $\times$ ____ = 1,680.

Puedo comprobar multiplicando ____ $\times$ ____ = 1,680.

Word bank: dividend divisor quotient partial quotients equally

Listen for: Students compute $1,680 \div 24 = 70$ and verify with $24 \times 70 = 1,680$, explaining the quotient is the per-classroom amount.

Level 2

If the school had 1,700 pencils instead, what would the quotient and remainder be, and what would the remainder mean in this situation?

- With 1,700 pencils the quotient is ____ and the remainder is ____.
- The remainder means ____ in real life because ____.

REFLECT

In a division problem, what is the difference between the quotient and the remainder, and what does each one tell you?

Level 1 support

Start here: The quotient is the size of each equal share; the remainder is what is left over.

 Need a hint?

Sentence starters (optional)

The quotient tells me ____, and the remainder tells me ____.

El cociente me dice ____, y el residuo me dice ____.

A remainder of 0 means ____.

Un residuo de 0 significa ____.

Word bank: quotient remainder dividend divisor divide

Listen for: Students explain the quotient is the number in each group while the remainder is the amount left that cannot be shared equally, and a remainder of 0 means it divides evenly.

Level 2

When is it correct to round a quotient UP because of a remainder, and when should you drop the remainder? Give an example of each.

- I round up when ____, for example ____.
- I drop the remainder when ____, for example ____.

Guided Examples

Example 1

What is $936 \div 12$?

Solution: $12 \times 78 = 936$. You can verify: $12 \times 70 = 840$, $12 \times 8 = 96$, $840 + 96 = 936$.

Answer: A. 78

Example 2

What is $2,485 \div 5$?

Solution: $5 \times 497 = 2,485$. Check: $5 \times 400 = 2,000$, $5 \times 90 = 450$, $5 \times 7 = 35$. $2,000 + 450 + 35 = 2,485$.

Answer: A. 497

Example 3

What is $756 \div 9$?

Solution: $9 \times 84 = 756$. Check: $9 \times 80 = 720$, $9 \times 4 = 36$, $720 + 36 = 756$.

Answer: A. 84

Write About the Math

The Writing Revolution

I can explain my division using the words dividend, divisor, quotient, and remainder.

1. Kernel Sentence subject + verb

Model: Partial quotients is a way to divide by breaking the problem into smaller, easier steps.
Cocientes parciales es una manera de dividir separando el problema en pasos más fáciles.

Write a kernel sentence about partial quotients. Use a subject and a verb.

Escribe una oración base sobre cocientes parciales. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Partial quotients matters in math
Cocientes parciales importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Partial quotients matters in math because ____.
Cocientes parciales importa en matemáticas porque ____.

but
pero

Partial quotients matters in math, but ____.
Cocientes parciales importa en matemáticas, pero ____.

so
entonces

Partial quotients matters in math, so ____.
Cocientes parciales importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about partial quotients.
Di un hecho verdadero sobre partial quotients.

Partial quotients ____.

Question *Pregunta*

Ask a question about partial quotients.
Haz una pregunta sobre partial quotients.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about partial quotients.
Muestra entusiasmo sobre partial quotients.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with partial quotients.
Dile a un compañero qué hacer con partial quotients.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Each section gets about ____ bars because 1,344 is close to ____.

Cada sección recibe unas ____ barras porque 1,344 está cerca de ____.

I estimated by ____.

Estimé al ____.

First I took $12 \times$ ____ = 1,200, then $12 \times$ ____ = 144.

Primero usé $12 \times$ ____ = 1,200, luego $12 \times$ ____ = 144.

Try It

Solve on your own. Check the answer key when you are done.

1. What is $1,125 \div 9$?

- A. 125
- B. 115
- C. 135
- D. 124

Show your work:

2. A warehouse has 2,184 items to pack into boxes of 14. How many boxes are needed?

- A. 156 boxes
- B. 146 boxes
- C. 166 boxes
- D. 155 boxes

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A school has 1,680 pencils to distribute equally to 24 classrooms. Show how to solve this using partial quotients. Explain each step and check your answer by multiplying.

Sentence starter: $1,680 \div 24 = \underline{\quad}$. First I used $24 \times \underline{\quad} = \underline{\quad}$, then $24 \times \underline{\quad} = \underline{\quad}$. The partial quotients add to $\underline{\quad}$. I checked by multiplying: $24 \times \underline{\quad} = 1,680$.

Show your work:

Reflect — Exit Ticket

What is $2,352 \div 16$?

- A. 147
- B. 137
- C. 157
- D. 146

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $125 - 9 \times 125 = 1,125$. *Check:* $9 \times 100 = 900$, $9 \times 25 = 225$, $900 + 225 = 1,125$.
2. **Try It 2:** A. 156 boxes — $2,184 \div 14 = 156$. *Check:* $14 \times 156 = 14 \times 150 + 14 \times 6 = 2,100 + 84 = 2,184$.
3. **Exit Ticket:** A. $147 - 16 \times 147 = 2,352$. *Check:* $16 \times 100 = 1,600$, $16 \times 40 = 640$, $16 \times 7 = 112$.
 $1,600 + 640 + 112 = 2,352$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Partial quotients is a way to divide by breaking the problem into smaller, easier steps.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).